Lesson notes · for you

Secret messages: encryption basics

Cyber security · Level 3 · Lesson 4

Time	Materials
20 min	Paper, a pen and scissors; Two strips of paper with the alphabet written on each

Learning goals

- Explain encryption, keys and decryption.
- Use Caesar's cipher, and crack it by trying keys or counting letters.
- Explain why modern encryption is strong, and where it protects you.

Activity: Cipher challenge

1. Make (4 min): Write the alphabet on two strips of paper. Slide one along the other to make any shift, and mark where A (or ا) lines up.
2. Hide (5 min): Choose a secret shift and encrypt a message of at least three sentences. Longer messages are easier to crack by counting, so try both a short and a long one.
3. Crack (8 min): Swap messages with a partner. First try shifts one by one; then count the letters and guess that the most common one stands for E, or alif.
4. Compare (3 min): Which method found the key faster? Why would neither work on the encryption in your chat app?

Arabic messages: use the 28 letters in order (ا ب ت ث ...), and write آ إ أ as ا, ؤ as و, ئ and ى as ي, and ة as ه, as the Labs app does.

Discussion

- Why do long messages make Caesar's cipher easier to crack?
- If the method is public, what keeps a message secret?
- Why do attackers trick people instead of breaking encryption?

Check yourself

- One place encryption protects me + one key I'll keep to myself.

Try it this week

Write a short message to a friend in Caesar's cipher and tell them the key in person. Then swap: each of you tries to crack the other's next message without the key, by trying shifts or counting letters. Time how long it takes.

Worksheet

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Name _____ Class _____ Date _____

1 **Why is Caesar's cipher easy to crack?**

◯ There are so few keys you can try them all ◯ Because it's very old

2 **Decrypt this message. It was written with Caesar's cipher and a shift of 3: KHOOR**

_____

3 **With end-to-end encryption, even the app company can't read your messages.**

◯ True ◯ False

4 **Match each word to its meaning.**

Plaintext	• • • •	The scrambled message
Ciphertext	• • • •	The readable message
Key	• • • •	Turning ciphertext back into plaintext
Decryption	• • • •	The secret that locks and unlocks it

5 **In which of these does encryption protect what you share?**

☐ A WhatsApp chat between you and a friend ☐ Photos on a phone with a screen lock

☐ Your message, after a friend forwards it to a group ☐ A password you told a classmate

☐ A page you open over HTTPS

✎ **Where does encryption already protect you every day? And where might you be giving away a key without thinking?**

_____

_____

_____

Check your answers

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1. **There are so few keys you can try them all**

   Age isn't the problem: some old ideas are still strong.

2. **hello**

   Shift each letter back by 3: K→H, H→E, O→L, O→L, R→O.

3. **True**

   With end-to-end encryption, only the people in the conversation hold the keys, so the app company can't read the messages either.

4. **Plaintext — The readable message**
   Ciphertext — The scrambled message
   Key — The secret that locks and unlocks it
   Decryption — Turning ciphertext back into plaintext

   Every cipher, from Caesar's to the one in your chat app, has these four parts.

5. **A WhatsApp chat between you and a friend; A page you open over HTTPS; Photos on a phone with a screen lock**

   Encryption protects the chat on its way, the connection to the site, and the phone's storage while it's locked.